Pace Analytical Services, Inc.
1638 Roseytown Road - Suites 2,3,4
Greensburg, PA 15601
(724)850-5600

January 28, 2016

Gene Gommer
Northstar
55 Progress Place 151
Jackson, NJ 08527

RE: Project: Transportation Drive 1546122
Pace Project No.: 30170621

Dear Gene Gommer:

Enclosed are the analytical results for sample(s) received by the laboratory on January 14, 2016. The results relate only to the samples included in this report. Results reported herein conform to the most current TNI standards and the laboratory's Quality Assurance Manual, where applicable, unless otherwise noted in the body of the report.

The samples were subcontracted to Pace Analytical Services, Inc., 1000 Riverbend Blvd., Suite F, St. Rose, LA 70087 for TCLP Herb & EOX analysis. Results of the analysis are reported on the Pace Analytical, New Orleans data tables.

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

Rachel D Christner

Rachel Christner
rachel.christner@pacelabs.com
Project Manager

Enclosures

Pace Analytical Services, Inc.
1638 Roseytown Road - Suites 2,3,4
Greensburg, PA 15601
(724)850-5600

# CERTIFICATIONS

| | |
|---|---|
| Project: | Transportation Drive 1546122 |
| Pace Project No.: | 30170621 |

**New Orleans Certification IDs**

California Env. Lab Accreditation Program Branch: 11277CA
Florida Department of Health (NELAC): E87595
Illinois Environmental Protection Agency: 0025721
Kansas Department of Health and Environment (NELAC): E-10266
Louisiana Dept. of Environmental Quality (NELAC/LELAP): 02006
Pennsylviania Dept. of Env Protection (NELAC): 68-04202
Texas Commission on Env. Quality (NELAC): T104704405-09-TX
U.S. Dept. of Agriculture Foreign Soil Import: P330-10-00119

**Pennsylvania Certification IDs**

Georgia Certification #: C040
1638 Roseytown Rd Suites 2,3&4, Greensburg, PA 15601
L-A-B DOD-ELAP Accreditation #: L2417
Alabama Certification #: 41590
Arizona Certification #: AZ0734
Arkansas Certification
California Certification #: 04222CA
Colorado Certification
Connecticut Certification #: PH-0694
Delaware Certification
Florida/TNI Certification #: E87683
Georgia Certification #: C040
Guam Certification
Hawaii Certification
Idaho Certification
Illinois Certification
Indiana Certification
Iowa Certification #: 391
Kansas/TNI Certification #: E-10358
Kentucky Certification #: 90133
Louisiana DHH/TNI Certification #: LA140008
Louisiana DEQ/TNI Certification #: 4086
Maine Certification #: PA00091
Maryland Certification #: 308
Massachusetts Certification #: M-PA1457
Michigan/PADEP Certification
Missouri Certification #: 235
Montana Certification #: Cert 0082
Nebraska Certification #: NE-05-29-14
Nevada Certification #: PA014572015-1
New Hampshire/TNI Certification #: 2976
New Jersey/TNI Certification #: PA 051
New Mexico Certification #: PA01457
New York/TNI Certification #: 10888
North Carolina Certification #: 42706
North Dakota Certification #: R-190
Oregon/TNI Certification #: PA200002
Pennsylvania/TNI Certification #: 65-00282
Puerto Rico Certification #: PA01457
Rhode Island Certification #: 65-00282
South Dakota Certification
Tennessee Certification #: TN2867
Texas/TNI Certification #: T104704188-14-8
Utah/TNI Certification #: PA014572015-5
USDA Soil Permit #: P330-14-00213
Vermont Dept. of Health: ID# VT-0282
Virgin Island/PADEP Certification
Virginia/VELAP Certification #: 460198
Washington Certification #: C868
West Virginia DEP Certification #: 143
West Virginia DHHR Certification #: 9964C
Wisconsin Certification
Wyoming Certification #: 8TMS-L

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## SAMPLE ANALYTE COUNT

Project: Transportation Drive 1546122
Pace Project No.: 30170621

| Lab ID | Sample ID | Method | Analysts | Analytes Reported | Laboratory |
|---|---|---|---|---|---|
| **30170621001** | **Schiavo SP-02-A** | EPA 8015B | CWB | 2 | PASI-PA |
| | | EPA 8081A | CWB | 8 | PASI-PA |
| | | EPA 8082 | SJG | 10 | PASI-PA |
| | | EPA 8151 | SLUC | 3 | PASI-N |
| | | EPA 6010B | CTS | 7 | PASI-PA |
| | | EPA 7470A | CTS | 1 | PASI-PA |
| | | EPA 9023 | DG | 1 | PASI-N |
| | | EPA 8270C | DJL | 18 | PASI-PA |
| | | EPA 8260B | LEL | 14 | PASI-PA |
| | | Dry Weight | SRA | 1 | PASI-PA |
| | | EPA 1010 | BMS | 1 | PASI-PA |
| | | EPA 9045C | LEP | 1 | PASI-PA |
| | | SW-846 7.3.3.2 | PAS | 1 | PASI-PA |
| | | SW-846 7.3.4.2 | PAS | 1 | PASI-PA |
| **30170621002** | **Schiavo SP-02-B** | EPA 8015B | CWB | 2 | PASI-PA |
| | | EPA 8081A | CWB | 8 | PASI-PA |
| | | EPA 8082 | SJG | 10 | PASI-PA |
| | | EPA 8151 | SLUC | 3 | PASI-N |
| | | EPA 6010B | CTS | 7 | PASI-PA |
| | | EPA 7470A | CTS | 1 | PASI-PA |
| | | EPA 9023 | DG | 1 | PASI-N |
| | | EPA 8270C | DJL | 18 | PASI-PA |
| | | EPA 8260B | LEL | 14 | PASI-PA |
| | | Dry Weight | SRA | 1 | PASI-PA |
| | | EPA 1010 | BMS | 1 | PASI-PA |
| | | EPA 9045C | LEP | 1 | PASI-PA |
| | | SW-846 7.3.3.2 | PAS | 1 | PASI-PA |
| | | SW-846 7.3.4.2 | PAS | 1 | PASI-PA |
| **30170621003** | **Schiavo SP-02-C** | EPA 8015B | CWB | 2 | PASI-PA |
| | | EPA 8081A | CWB | 8 | PASI-PA |
| | | EPA 8082 | SJG | 10 | PASI-PA |
| | | EPA 8151 | SLUC | 3 | PASI-N |
| | | EPA 6010B | CTS | 7 | PASI-PA |
| | | EPA 7470A | CTS | 1 | PASI-PA |
| | | EPA 9023 | DG | 1 | PASI-N |
| | | EPA 8270C | DJL | 18 | PASI-PA |
| | | EPA 8260B | LEL | 14 | PASI-PA |

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## SAMPLE ANALYTE COUNT

Project: Transportation Drive 1546122
Pace Project No.: 30170621

| Lab ID | Sample ID | Method | Analysts | Analytes Reported | Laboratory |
|---|---|---|---|---|---|
| | | Dry Weight | SRA | 1 | PASI-PA |
| | | EPA 1010 | BMS | 1 | PASI-PA |
| | | EPA 9045C | LEP | 1 | PASI-PA |
| | | SW-846 7.3.3.2 | PAS | 1 | PASI-PA |
| | | SW-846 7.3.4.2 | PAS | 1 | PASI-PA |

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## ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-A** **Lab ID: 30170621001** Collected: 01/13/16 10:00 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8015 TPH Microwave** | Analytical Method: EPA 8015B Preparation Method: EPA 3546 | | | | | | | |
| TPH (C10-C28) | **573** | mg/kg | 78.3 | 10 | 01/21/16 10:20 | 01/22/16 03:19 | | |
| ***Surrogates*** | | | | | | | | |
| o-Terphenyl (S) | 468 | % | 45-103 | 10 | 01/21/16 10:20 | 01/22/16 03:19 | 84-15-1 | S4 |
| **8081 GCS Pesticides, TCLP** | Analytical Method: EPA 8081A Preparation Method: EPA 3510C | | | | | | | |
| | Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 7.8; Final pH: 5.01 | | | | | | | |
| gamma-BHC (Lindane) | ND | ug/L | 10.0 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 58-89-9 | 1c |
| Chlordane (Technical) | ND | ug/L | 10.0 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 57-74-9 | 1c |
| Endrin | ND | ug/L | 1.0 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 72-20-8 | 1c |
| Heptachlor epoxide | ND | ug/L | 0.50 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 1024-57-3 | 1c |
| Methoxychlor | ND | ug/L | 100 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 72-43-5 | 1c |
| Toxaphene | ND | ug/L | 50.0 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 8001-35-2 | 1c |
| ***Surrogates*** | | | | | | | | |
| Decachlorobiphenyl (S) | 36 | % | 15-125 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 2051-24-3 | 1c,SS |
| Tetrachloro-m-xylene (S) | 57 | % | 14-136 | 1 | 01/20/16 14:45 | 01/21/16 16:42 | 877-09-8 | 1c |
| **8082 GCS PCB** | Analytical Method: EPA 8082 Preparation Method: EPA 3546 | | | | | | | |
| PCB-1016 (Aroclor 1016) | ND | ug/kg | 9770 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 12674-11-2 | |
| PCB-1221 (Aroclor 1221) | ND | ug/kg | 9770 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 11104-28-2 | |
| PCB-1232 (Aroclor 1232) | ND | ug/kg | 9770 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 11141-16-5 | |
| PCB-1242 (Aroclor 1242) | ND | ug/kg | 9770 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 53469-21-9 | |
| PCB-1248 (Aroclor 1248) | **54700** | ug/kg | 9770 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 12672-29-6 | |
| PCB-1254 (Aroclor 1254) | **64400** | ug/kg | 9770 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 11097-69-1 | 3c |
| PCB-1260 (Aroclor 1260) | ND | ug/kg | 9770 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 11096-82-5 | |
| PCB, Total | **119000** | ug/kg | 68400 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 1336-36-3 | |
| ***Surrogates*** | | | | | | | | |
| Tetrachloro-m-xylene (S) | 121 | % | 30-107 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 877-09-8 | CH,S4 |
| Decachlorobiphenyl (S) | 187 | % | 10-115 | 500 | 01/19/16 17:09 | 01/26/16 07:16 | 2051-24-3 | CH,S4 |
| **8151A Cl Herbicides TCLP** | Analytical Method: EPA 8151 Preparation Method: EPA 3535A | | | | | | | |
| | Leachate Method/Date: EPA 1311; 01/18/16 15:30 | | | | | | | |
| 2,4-D | ND | mg/L | 0.020 | 1 | 01/20/16 14:27 | 01/22/16 15:19 | 94-75-7 | R1 |
| 2,4,5-TP (Silvex) | ND | mg/L | 0.020 | 1 | 01/20/16 14:27 | 01/22/16 15:19 | 93-72-1 | R1 |
| ***Surrogates*** | | | | | | | | |
| 2,4-DCAA (S) | 76 | %. | 10-166 | 1 | 01/20/16 14:27 | 01/22/16 15:19 | 19719-28-9 | |
| 2,4-DCAA (S) | 77 | %. | 10-166 | 1 | 01/20/16 14:27 | 01/22/16 15:19 | 19719-28-9 | |
| **6010 MET ICP, TCLP** | Analytical Method: EPA 6010B Preparation Method: EPA 3005A | | | | | | | |
| | Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 7.8; Final pH: 5.01 | | | | | | | |
| Arsenic | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:39 | 7440-38-2 | |
| Barium | **1.6** | mg/L | 1.0 | 1 | 01/19/16 13:25 | 01/20/16 08:39 | 7440-39-3 | |
| Cadmium | **0.087** | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:39 | 7440-43-9 | |
| Chromium | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:39 | 7440-47-3 | |
| Lead | **13.7** | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:39 | 7439-92-1 | |
| Selenium | ND | mg/L | 0.10 | 1 | 01/19/16 13:25 | 01/20/16 08:39 | 7782-49-2 | |
| Silver | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:39 | 7440-22-4 | |

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# ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-A** **Lab ID: 30170621001** Collected: 01/13/16 10:00 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **7470 Mercury, TCLP** | Analytical Method: EPA 7470A Preparation Method: EPA 7470A | | | | | | | |
| | Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 7.8; Final pH: 5.01 | | | | | | | |
| Mercury | ND | ug/L | 1.0 | 1 | 01/19/16 17:08 | 01/20/16 09:06 | 7439-97-6 | |
| **9023 Ext. Organic Halides EOX** | Analytical Method: EPA 9023 Preparation Method: EPA 9023 | | | | | | | |
| Extractable Organic Halogens | ND | mg/kg | 46.7 | 1 | 01/27/16 10:06 | 01/27/16 11:09 | | |
| **8270 MSSV TCLP Sep Funnel** | Analytical Method: EPA 8270C Preparation Method: EPA 3510C | | | | | | | |
| 1,4-Dichlorobenzene | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 106-46-7 | |
| 2,4-Dinitrotoluene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 121-14-2 | |
| Hexachloro-1,3-butadiene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 87-68-3 | |
| Hexachlorobenzene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 118-74-1 | |
| Hexachloroethane | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 67-72-1 | |
| 2-Methylphenol(o-Cresol) | ND | ug/L | 2000 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 95-48-7 | |
| 3&4-Methylphenol(m&p Cresol) | ND | ug/L | 2000 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | | |
| Nitrobenzene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 98-95-3 | |
| Pentachlorophenol | ND | ug/L | 5000 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 87-86-5 | |
| Pyridine | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 110-86-1 | |
| 2,4,5-Trichlorophenol | ND | ug/L | 5000 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 95-95-4 | |
| 2,4,6-Trichlorophenol | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 88-06-2 | |
| ***Surrogates*** | | | | | | | | |
| Nitrobenzene-d5 (S) | 78 | % | 22-128 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 4165-60-0 | |
| 2-Fluorobiphenyl (S) | 76 | % | 34-113 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 321-60-8 | |
| Terphenyl-d14 (S) | 154 | % | 35-150 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 1718-51-0 | IS,S3 |
| Phenol-d6 (S) | 39 | % | 14-49 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 13127-88-3 | |
| 2-Fluorophenol (S) | 58 | % | 19-70 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 367-12-4 | |
| 2,4,6-Tribromophenol (S) | 102 | % | 34-134 | 1 | 01/19/16 12:35 | 01/20/16 00:50 | 118-79-6 | |
| **8260B MSV TCLP** | Analytical Method: EPA 8260B Leachate Method/Date: EPA 1311; 01/19/16 17:00 | | | | | | | |
| Benzene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:33 | 71-43-2 | |
| 2-Butanone (MEK) | ND | ug/L | 5000 | 10 | | 01/26/16 08:33 | 78-93-3 | |
| Carbon tetrachloride | ND | ug/L | 50.0 | 10 | | 01/26/16 08:33 | 56-23-5 | |
| Chlorobenzene | ND | ug/L | 1000 | 10 | | 01/26/16 08:33 | 108-90-7 | M1 |
| Chloroform | ND | ug/L | 500 | 10 | | 01/26/16 08:33 | 67-66-3 | |
| 1,2-Dichloroethane | ND | ug/L | 50.0 | 10 | | 01/26/16 08:33 | 107-06-2 | |
| 1,1-Dichloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:33 | 75-35-4 | |
| Tetrachloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:33 | 127-18-4 | |
| Trichloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:33 | 79-01-6 | |
| Vinyl chloride | ND | ug/L | 50.0 | 10 | | 01/26/16 08:33 | 75-01-4 | |
| ***Surrogates*** | | | | | | | | |
| 1,2-Dichloroethane-d4 (S) | 98 | % | 77-126 | 10 | | 01/26/16 08:33 | 17060-07-0 | |
| Toluene-d8 (S) | 89 | % | 84-115 | 10 | | 01/26/16 08:33 | 2037-26-5 | |
| 4-Bromofluorobenzene (S) | 89 | % | 81-119 | 10 | | 01/26/16 08:33 | 460-00-4 | |
| Dibromofluoromethane (S) | 97 | % | 70-130 | 10 | | 01/26/16 08:33 | 1868-53-7 | |

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## ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-A** **Lab ID: 30170621001** Collected: 01/13/16 10:00 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **Percent Moisture** | Analytical Method: Dry Weight | | | | | | | |
| Percent Moisture | **16.8** | % | 0.10 | 1 | | 01/21/16 18:21 | | |
| **1010 Flashpoint,Closed Cup** | Analytical Method: EPA 1010 | | | | | | | |
| Flashpoint | **>200** | deg F | 60.0 | 1 | | 01/22/16 16:58 | | |
| **9045 pH Soil** | Analytical Method: EPA 9045C | | | | | | | |
| pH at 25 Degrees C | **6.4** | Std. Units | 1.0 | 1 | | 01/15/16 21:07 | | |
| **733C S Reactive Cyanide** | Analytical Method: SW-846 7.3.3.2 | | | | | | | |
| Cyanide, Reactive | ND | mg/kg | 1.2 | 1 | | 01/19/16 23:38 | | |
| **735S Reactive Sulfide** | Analytical Method: SW-846 7.3.4.2 | | | | | | | |
| Sulfide, Reactive | ND | mg/kg | 11.9 | 1 | | 01/19/16 16:25 | | |

**Sample: Schiavo SP-02-B** **Lab ID: 30170621002** Collected: 01/13/16 10:45 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8015 TPH Microwave** | Analytical Method: EPA 8015B Preparation Method: EPA 3546 | | | | | | | |
| TPH (C10-C28) | **192** | mg/kg | 16.0 | 2 | 01/21/16 10:20 | 01/22/16 05:46 | | |
| ***Surrogates*** | | | | | | | | |
| o-Terphenyl (S) | 66 | % | 45-103 | 2 | 01/21/16 10:20 | 01/22/16 05:46 | 84-15-1 | |
| **8081 GCS Pesticides, TCLP** | Analytical Method: EPA 8081A Preparation Method: EPA 3510C | | | | | | | |
| | Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 7.65; Final pH: 4.97 | | | | | | | |
| gamma-BHC (Lindane) | ND | ug/L | 10.0 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 58-89-9 | 1c |
| Chlordane (Technical) | ND | ug/L | 10.0 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 57-74-9 | 1c |
| Endrin | ND | ug/L | 1.0 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 72-20-8 | 1c |
| Heptachlor epoxide | ND | ug/L | 0.50 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 1024-57-3 | 1c |
| Methoxychlor | ND | ug/L | 100 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 72-43-5 | 1c |
| Toxaphene | ND | ug/L | 50.0 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 8001-35-2 | 1c |
| ***Surrogates*** | | | | | | | | |
| Decachlorobiphenyl (S) | 51 | % | 15-125 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 2051-24-3 | 1c,SS |
| Tetrachloro-m-xylene (S) | 59 | % | 14-136 | 1 | 01/20/16 14:45 | 01/21/16 17:09 | 877-09-8 | 1c |
| **8082 GCS PCB** | Analytical Method: EPA 8082 Preparation Method: EPA 3546 | | | | | | | |
| PCB-1016 (Aroclor 1016) | ND | ug/kg | 9990 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 12674-11-2 | |
| PCB-1221 (Aroclor 1221) | ND | ug/kg | 9990 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 11104-28-2 | |
| PCB-1232 (Aroclor 1232) | ND | ug/kg | 9990 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 11141-16-5 | |
| PCB-1242 (Aroclor 1242) | ND | ug/kg | 9990 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 53469-21-9 | |
| PCB-1248 (Aroclor 1248) | **53200** | ug/kg | 9990 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 12672-29-6 | |
| PCB-1254 (Aroclor 1254) | **97500** | ug/kg | 9990 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 11097-69-1 | 3c |
| PCB-1260 (Aroclor 1260) | ND | ug/kg | 9990 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 11096-82-5 | |

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# ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-B** **Lab ID: 30170621002** Collected: 01/13/16 10:45 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8082 GCS PCB** | Analytical Method: EPA 8082 Preparation Method: EPA 3546 | | | | | | | |
| PCB, Total | **151000** | ug/kg | 70000 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 1336-36-3 | |
| ***Surrogates*** | | | | | | | | |
| Tetrachloro-m-xylene (S) | 115 | % | 30-107 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 877-09-8 | CH,S4 |
| Decachlorobiphenyl (S) | 116 | % | 10-115 | 500 | 01/19/16 17:09 | 01/26/16 07:36 | 2051-24-3 | CH,S4 |
| **8151A Cl Herbicides TCLP** | Analytical Method: EPA 8151 Preparation Method: EPA 3535A<br>Leachate Method/Date: EPA 1311; 01/18/16 15:30 | | | | | | | |
| 2,4-D | ND | mg/L | 0.020 | 1 | 01/20/16 14:27 | 01/22/16 14:14 | 94-75-7 | |
| 2,4,5-TP (Silvex) | ND | mg/L | 0.020 | 1 | 01/20/16 14:27 | 01/22/16 14:14 | 93-72-1 | |
| ***Surrogates*** | | | | | | | | |
| 2,4-DCAA (S) | 86 | %. | 10-166 | 1 | 01/20/16 14:27 | 01/22/16 14:14 | 19719-28-9 | |
| 2,4-DCAA (S) | 83 | %. | 10-166 | 1 | 01/20/16 14:27 | 01/22/16 14:14 | 19719-28-9 | |
| **6010 MET ICP, TCLP** | Analytical Method: EPA 6010B Preparation Method: EPA 3005A<br>Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 7.65; Final pH: 4.97 | | | | | | | |
| Arsenic | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:41 | 7440-38-2 | |
| Barium | **1.5** | mg/L | 1.0 | 1 | 01/19/16 13:25 | 01/20/16 08:41 | 7440-39-3 | |
| Cadmium | **0.055** | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:41 | 7440-43-9 | |
| Chromium | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:41 | 7440-47-3 | |
| Lead | **1.6** | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:41 | 7439-92-1 | |
| Selenium | ND | mg/L | 0.10 | 1 | 01/19/16 13:25 | 01/20/16 08:41 | 7782-49-2 | |
| Silver | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:41 | 7440-22-4 | |
| **7470 Mercury, TCLP** | Analytical Method: EPA 7470A Preparation Method: EPA 7470A<br>Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 7.65; Final pH: 4.97 | | | | | | | |
| Mercury | ND | ug/L | 1.0 | 1 | 01/19/16 17:08 | 01/20/16 09:11 | 7439-97-6 | |
| **9023 Ext. Organic Halides EOX** | Analytical Method: EPA 9023 Preparation Method: EPA 9023 | | | | | | | |
| Extractable Organic Halogens | ND | mg/kg | 46.1 | 1 | 01/27/16 10:06 | 01/27/16 11:48 | | |
| **8270 MSSV TCLP Sep Funnel** | Analytical Method: EPA 8270C Preparation Method: EPA 3510C | | | | | | | |
| 1,4-Dichlorobenzene | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 106-46-7 | |
| 2,4-Dinitrotoluene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 121-14-2 | |
| Hexachloro-1,3-butadiene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 87-68-3 | |
| Hexachlorobenzene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 118-74-1 | |
| Hexachloroethane | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 67-72-1 | |
| 2-Methylphenol(o-Cresol) | ND | ug/L | 2000 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 95-48-7 | |
| 3&4-Methylphenol(m&p Cresol) | ND | ug/L | 2000 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | | |
| Nitrobenzene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 98-95-3 | |
| Pentachlorophenol | ND | ug/L | 5000 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 87-86-5 | |
| Pyridine | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 110-86-1 | |
| 2,4,5-Trichlorophenol | ND | ug/L | 5000 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 95-95-4 | |
| 2,4,6-Trichlorophenol | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 88-06-2 | |
| ***Surrogates*** | | | | | | | | |
| Nitrobenzene-d5 (S) | 79 | % | 22-128 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 4165-60-0 | |
| 2-Fluorobiphenyl (S) | 79 | % | 34-113 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 321-60-8 | |

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# ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-B** **Lab ID: 30170621002** Collected: 01/13/16 10:45 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8270 MSSV TCLP Sep Funnel** | Analytical Method: EPA 8270C Preparation Method: EPA 3510C | | | | | | | |
| ***Surrogates*** | | | | | | | | |
| Terphenyl-d14 (S) | 151 | % | 35-150 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 1718-51-0 | S3 |
| Phenol-d6 (S) | 40 | % | 14-49 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 13127-88-3 | |
| 2-Fluorophenol (S) | 59 | % | 19-70 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 367-12-4 | |
| 2,4,6-Tribromophenol (S) | 115 | % | 34-134 | 1 | 01/19/16 12:35 | 01/20/16 01:09 | 118-79-6 | |
| **8260B MSV TCLP** | Analytical Method: EPA 8260B Leachate Method/Date: EPA 1311; 01/19/16 17:00 | | | | | | | |
| Benzene | ND | ug/L | 50.0 | 10 | | 01/26/16 07:40 | 71-43-2 | |
| 2-Butanone (MEK) | ND | ug/L | 5000 | 10 | | 01/26/16 07:40 | 78-93-3 | |
| Carbon tetrachloride | ND | ug/L | 50.0 | 10 | | 01/26/16 07:40 | 56-23-5 | |
| Chlorobenzene | ND | ug/L | 1000 | 10 | | 01/26/16 07:40 | 108-90-7 | |
| Chloroform | ND | ug/L | 500 | 10 | | 01/26/16 07:40 | 67-66-3 | |
| 1,2-Dichloroethane | ND | ug/L | 50.0 | 10 | | 01/26/16 07:40 | 107-06-2 | |
| 1,1-Dichloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 07:40 | 75-35-4 | |
| Tetrachloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 07:40 | 127-18-4 | |
| Trichloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 07:40 | 79-01-6 | |
| Vinyl chloride | ND | ug/L | 50.0 | 10 | | 01/26/16 07:40 | 75-01-4 | |
| ***Surrogates*** | | | | | | | | |
| 1,2-Dichloroethane-d4 (S) | 100 | % | 77-126 | 10 | | 01/26/16 07:40 | 17060-07-0 | |
| Toluene-d8 (S) | 90 | % | 84-115 | 10 | | 01/26/16 07:40 | 2037-26-5 | |
| 4-Bromofluorobenzene (S) | 92 | % | 81-119 | 10 | | 01/26/16 07:40 | 460-00-4 | |
| Dibromofluoromethane (S) | 96 | % | 70-130 | 10 | | 01/26/16 07:40 | 1868-53-7 | |
| **Percent Moisture** | Analytical Method: Dry Weight | | | | | | | |
| Percent Moisture | **17.9** | % | 0.10 | 1 | | 01/21/16 18:21 | | |
| **1010 Flashpoint,Closed Cup** | Analytical Method: EPA 1010 | | | | | | | |
| Flashpoint | **>200** | deg F | 60.0 | 1 | | 01/22/16 16:58 | | |
| **9045 pH Soil** | Analytical Method: EPA 9045C | | | | | | | |
| pH at 25 Degrees C | **6.2** | Std. Units | 1.0 | 1 | | 01/15/16 21:07 | | |
| **733C S Reactive Cyanide** | Analytical Method: SW-846 7.3.3.2 | | | | | | | |
| Cyanide, Reactive | ND | mg/kg | 1.2 | 1 | | 01/19/16 23:39 | | |
| **735S Reactive Sulfide** | Analytical Method: SW-846 7.3.4.2 | | | | | | | |
| Sulfide, Reactive | ND | mg/kg | 12.1 | 1 | | 01/19/16 16:25 | | |

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## ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-C** **Lab ID: 30170621003** Collected: 01/13/16 11:45 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **8015 TPH Microwave** | Analytical Method: EPA 8015B Preparation Method: EPA 3546 | | | | | | | |
| TPH (C10-C28) | **184** | mg/kg | 16.1 | 2 | 01/21/16 10:20 | 01/22/16 06:20 | | |
| ***Surrogates*** | | | | | | | | |
| o-Terphenyl (S) | 62 | % | 45-103 | 2 | 01/21/16 10:20 | 01/22/16 06:20 | 84-15-1 | |
| **8081 GCS Pesticides, TCLP** | Analytical Method: EPA 8081A Preparation Method: EPA 3510C<br>Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 6.98; Final pH: 4.96 | | | | | | | |
| gamma-BHC (Lindane) | ND | ug/L | 10.0 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 58-89-9 | 1c |
| Chlordane (Technical) | ND | ug/L | 10.0 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 57-74-9 | 1c |
| Endrin | ND | ug/L | 1.0 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 72-20-8 | 1c |
| Heptachlor epoxide | ND | ug/L | 0.50 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 1024-57-3 | 1c |
| Methoxychlor | ND | ug/L | 100 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 72-43-5 | 1c |
| Toxaphene | ND | ug/L | 50.0 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 8001-35-2 | 1c |
| ***Surrogates*** | | | | | | | | |
| Decachlorobiphenyl (S) | 50 | % | 15-125 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 2051-24-3 | 1c,SS |
| Tetrachloro-m-xylene (S) | 57 | % | 14-136 | 1 | 01/20/16 14:45 | 01/21/16 17:37 | 877-09-8 | 1c |
| **8082 GCS PCB** | Analytical Method: EPA 8082 Preparation Method: EPA 3546 | | | | | | | |
| PCB-1016 (Aroclor 1016) | ND | ug/kg | 5060 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 12674-11-2 | |
| PCB-1221 (Aroclor 1221) | ND | ug/kg | 5060 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 11104-28-2 | |
| PCB-1232 (Aroclor 1232) | ND | ug/kg | 5060 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 11141-16-5 | |
| PCB-1242 (Aroclor 1242) | ND | ug/kg | 5060 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 53469-21-9 | |
| PCB-1248 (Aroclor 1248) | **14400** | ug/kg | 5060 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 12672-29-6 | 2c |
| PCB-1254 (Aroclor 1254) | **24100** | ug/kg | 5060 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 11097-69-1 | 2c |
| PCB-1260 (Aroclor 1260) | ND | ug/kg | 5060 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 11096-82-5 | |
| PCB, Total | **38500** | ug/kg | 35400 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 1336-36-3 | |
| ***Surrogates*** | | | | | | | | |
| Tetrachloro-m-xylene (S) | 66 | % | 30-107 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 877-09-8 | |
| Decachlorobiphenyl (S) | 102 | % | 10-115 | 250 | 01/19/16 17:09 | 01/26/16 13:22 | 2051-24-3 | CL |
| **8151A Cl Herbicides TCLP** | Analytical Method: EPA 8151 Preparation Method: EPA 3535A<br>Leachate Method/Date: EPA 1311; 01/18/16 15:30 | | | | | | | |
| 2,4-D | ND | mg/L | 0.020 | 1 | 01/20/16 14:27 | 01/22/16 15:03 | 94-75-7 | |
| 2,4,5-TP (Silvex) | ND | mg/L | 0.020 | 1 | 01/20/16 14:27 | 01/22/16 15:03 | 93-72-1 | |
| ***Surrogates*** | | | | | | | | |
| 2,4-DCAA (S) | 77 | %. | 10-166 | 1 | 01/20/16 14:27 | 01/22/16 15:03 | 19719-28-9 | |
| 2,4-DCAA (S) | 74 | %. | 10-166 | 1 | 01/20/16 14:27 | 01/22/16 15:03 | 19719-28-9 | |
| **6010 MET ICP, TCLP** | Analytical Method: EPA 6010B Preparation Method: EPA 3005A<br>Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 6.98; Final pH: 4.96 | | | | | | | |
| Arsenic | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:44 | 7440-38-2 | |
| Barium | **1.2** | mg/L | 1.0 | 1 | 01/19/16 13:25 | 01/20/16 08:44 | 7440-39-3 | |
| Cadmium | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:44 | 7440-43-9 | |
| Chromium | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:44 | 7440-47-3 | |
| Lead | **1.3** | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:44 | 7439-92-1 | |
| Selenium | ND | mg/L | 0.10 | 1 | 01/19/16 13:25 | 01/20/16 08:44 | 7782-49-2 | |
| Silver | ND | mg/L | 0.050 | 1 | 01/19/16 13:25 | 01/20/16 08:44 | 7440-22-4 | |

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# ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-C** **Lab ID: 30170621003** Collected: 01/13/16 11:45 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **7470 Mercury, TCLP** | Analytical Method: EPA 7470A Preparation Method: EPA 7470A<br>Leachate Method/Date: EPA 1311; 01/18/16 16:00 Initial pH: 6.98; Final pH: 4.96 | | | | | | | |
| Mercury | ND | ug/L | 1.0 | 1 | 01/19/16 17:08 | 01/20/16 09:12 | 7439-97-6 | |
| **9023 Ext. Organic Halides EOX** | Analytical Method: EPA 9023 Preparation Method: EPA 9023 | | | | | | | |
| Extractable Organic Halogens | ND | mg/kg | 49.3 | 1 | 01/27/16 10:06 | 01/27/16 12:03 | | |
| **8270 MSSV TCLP Sep Funnel** | Analytical Method: EPA 8270C Preparation Method: EPA 3510C | | | | | | | |
| 1,4-Dichlorobenzene | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 106-46-7 | |
| 2,4-Dinitrotoluene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 121-14-2 | |
| Hexachloro-1,3-butadiene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 87-68-3 | |
| Hexachlorobenzene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 118-74-1 | |
| Hexachloroethane | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 67-72-1 | |
| 2-Methylphenol(o-Cresol) | ND | ug/L | 2000 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 95-48-7 | |
| 3&4-Methylphenol(m&p Cresol) | ND | ug/L | 2000 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | | |
| Nitrobenzene | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 98-95-3 | |
| Pentachlorophenol | ND | ug/L | 5000 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 87-86-5 | |
| Pyridine | ND | ug/L | 500 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 110-86-1 | |
| 2,4,5-Trichlorophenol | ND | ug/L | 5000 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 95-95-4 | |
| 2,4,6-Trichlorophenol | ND | ug/L | 100 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 88-06-2 | |
| ***Surrogates*** | | | | | | | | |
| Nitrobenzene-d5 (S) | 82 | % | 22-128 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 4165-60-0 | |
| 2-Fluorobiphenyl (S) | 78 | % | 34-113 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 321-60-8 | |
| Terphenyl-d14 (S) | 145 | % | 35-150 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 1718-51-0 | |
| Phenol-d6 (S) | 40 | % | 14-49 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 13127-88-3 | |
| 2-Fluorophenol (S) | 58 | % | 19-70 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 367-12-4 | |
| 2,4,6-Tribromophenol (S) | 107 | % | 34-134 | 1 | 01/19/16 12:35 | 01/20/16 01:27 | 118-79-6 | |
| **8260B MSV TCLP** | Analytical Method: EPA 8260B Leachate Method/Date: EPA 1311; 01/19/16 17:00 | | | | | | | |
| Benzene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:06 | 71-43-2 | |
| 2-Butanone (MEK) | ND | ug/L | 5000 | 10 | | 01/26/16 08:06 | 78-93-3 | |
| Carbon tetrachloride | ND | ug/L | 50.0 | 10 | | 01/26/16 08:06 | 56-23-5 | |
| Chlorobenzene | ND | ug/L | 1000 | 10 | | 01/26/16 08:06 | 108-90-7 | |
| Chloroform | ND | ug/L | 500 | 10 | | 01/26/16 08:06 | 67-66-3 | |
| 1,2-Dichloroethane | ND | ug/L | 50.0 | 10 | | 01/26/16 08:06 | 107-06-2 | |
| 1,1-Dichloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:06 | 75-35-4 | |
| Tetrachloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:06 | 127-18-4 | |
| Trichloroethene | ND | ug/L | 50.0 | 10 | | 01/26/16 08:06 | 79-01-6 | |
| Vinyl chloride | ND | ug/L | 50.0 | 10 | | 01/26/16 08:06 | 75-01-4 | |
| ***Surrogates*** | | | | | | | | |
| 1,2-Dichloroethane-d4 (S) | 98 | % | 77-126 | 10 | | 01/26/16 08:06 | 17060-07-0 | |
| Toluene-d8 (S) | 94 | % | 84-115 | 10 | | 01/26/16 08:06 | 2037-26-5 | |
| 4-Bromofluorobenzene (S) | 96 | % | 81-119 | 10 | | 01/26/16 08:06 | 460-00-4 | |
| Dibromofluoromethane (S) | 91 | % | 70-130 | 10 | | 01/26/16 08:06 | 1868-53-7 | |

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## ANALYTICAL RESULTS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

**Sample: Schiavo SP-02-C** **Lab ID: 30170621003** Collected: 01/13/16 11:45 Received: 01/14/16 22:30 Matrix: Solid

***Results reported on a "dry weight" basis and are adjusted for percent moisture, sample size and any dilutions.***

| Parameters | Results | Units | Report Limit | DF | Prepared | Analyzed | CAS No. | Qual |
|---|---|---|---|---|---|---|---|---|
| **Percent Moisture** | Analytical Method: Dry Weight | | | | | | | |
| Percent Moisture | **19.8** | % | 0.10 | 1 | | 01/21/16 18:22 | | |
| **1010 Flashpoint,Closed Cup** | Analytical Method: EPA 1010 | | | | | | | |
| Flashpoint | **>200** | deg F | 60.0 | 1 | | 01/22/16 16:58 | | |
| **9045 pH Soil** | Analytical Method: EPA 9045C | | | | | | | |
| pH at 25 Degrees C | **6.2** | Std. Units | 1.0 | 1 | | 01/15/16 21:07 | | |
| **733C S Reactive Cyanide** | Analytical Method: SW-846 7.3.3.2 | | | | | | | |
| Cyanide, Reactive | ND | mg/kg | 1.2 | 1 | | 01/19/16 23:41 | | |
| **735S Reactive Sulfide** | Analytical Method: SW-846 7.3.4.2 | | | | | | | |
| Sulfide, Reactive | ND | mg/kg | 12.4 | 1 | | 01/19/16 16:25 | | |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: MERP/7283 | Analysis Method: EPA 7470A
QC Batch Method: EPA 7470A | Analysis Description: 7470 Mercury TCLP
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013299 | Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Mercury | ug/L | ND | 1.0 | 01/20/16 08:51 | |

LABORATORY CONTROL SAMPLE: 1013300

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Mercury | ug/L | 1 | 1.1 | 108 | 85-115 | |

MATRIX SPIKE SAMPLE: 1013302

| Parameter | Units | 30170669001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | ug/L | ND | 2.5 | 2.4 | 97 | 80-120 | |

MATRIX SPIKE SAMPLE: 1013304

| Parameter | Units | 30170651002 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Mercury | ug/L | ND | 2.5 | 2.5 | 102 | 80-120 | |

SAMPLE DUPLICATE: 1013301

| Parameter | Units | 30170669001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Mercury | ug/L | ND | ND | | |

SAMPLE DUPLICATE: 1013303

| Parameter | Units | 30170651002 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Mercury | ug/L | ND | ND | | |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: MPRP/17380 | Analysis Method: EPA 6010B
QC Batch Method: EPA 3005A | Analysis Description: 6010 MET TCLP
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013395 | Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Arsenic | mg/L | ND | 0.050 | 01/20/16 08:09 | |
| Barium | mg/L | ND | 1.0 | 01/20/16 08:09 | |
| Cadmium | mg/L | ND | 0.050 | 01/20/16 08:09 | |
| Chromium | mg/L | ND | 0.050 | 01/20/16 08:09 | |
| Lead | mg/L | ND | 0.050 | 01/20/16 08:09 | |
| Selenium | mg/L | ND | 0.10 | 01/20/16 08:09 | |
| Silver | mg/L | ND | 0.050 | 01/20/16 08:09 | |

LABORATORY CONTROL SAMPLE: 1013396

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Arsenic | mg/L | .5 | 0.51 | 102 | 80-120 | |
| Barium | mg/L | .5 | .55J | 109 | 80-120 | |
| Cadmium | mg/L | .5 | 0.53 | 105 | 80-120 | |
| Chromium | mg/L | .5 | 0.52 | 103 | 80-120 | |
| Lead | mg/L | .5 | 0.51 | 101 | 80-120 | |
| Selenium | mg/L | .5 | 0.52 | 105 | 80-120 | |
| Silver | mg/L | .25 | 0.25 | 100 | 80-120 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 1013398 | 1013399

| Parameter | Units | 30170651002 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | .5 | .5 | 0.59 | 0.57 | 115 | 112 | 75-125 | 3 | |
| Barium | mg/L | ND | .5 | .5 | .65J | .62J | 106 | 101 | 75-125 | | |
| Cadmium | mg/L | ND | .5 | .5 | 0.60 | 0.58 | 119 | 117 | 75-125 | 2 | |
| Chromium | mg/L | ND | .5 | .5 | 0.51 | 0.50 | 102 | 100 | 75-125 | 2 | |
| Lead | mg/L | ND | .5 | .5 | 0.50 | 0.49 | 99 | 97 | 75-125 | 2 | |
| Selenium | mg/L | ND | .5 | .5 | 0.60 | 0.59 | 119 | 116 | 75-125 | 3 | |
| Silver | mg/L | ND | .25 | .25 | 0.32 | 0.30 | 128 | 121 | 75-125 | 6 | M1 |

MATRIX SPIKE SAMPLE: 1013402

| Parameter | Units | 30170669001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Arsenic | mg/L | ND | .5 | 0.58 | 113 | 75-125 | |
| Barium | mg/L | ND | .5 | .79J | 101 | 75-125 | |
| Cadmium | mg/L | ND | .5 | 0.59 | 117 | 75-125 | |

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## QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

MATRIX SPIKE SAMPLE: 1013402

| Parameter | Units | 30170669001 Result | Spike Conc. | MS Result | MS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|---|
| Chromium | mg/L | ND | .5 | 0.49 | 97 | 75-125 | |
| Lead | mg/L | ND | .5 | 0.49 | 95 | 75-125 | |
| Selenium | mg/L | ND | .5 | 0.60 | 118 | 75-125 | |
| Silver | mg/L | ND | .25 | 0.31 | 124 | 75-125 | |

SAMPLE DUPLICATE: 1013397

| Parameter | Units | 30170651002 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Arsenic | mg/L | ND | .0093J | | |
| Barium | mg/L | ND | .12J | | |
| Cadmium | mg/L | ND | ND | | |
| Chromium | mg/L | ND | .0014J | | |
| Lead | mg/L | ND | ND | | |
| Selenium | mg/L | ND | .0071J | | |
| Silver | mg/L | ND | ND | | |

SAMPLE DUPLICATE: 1013401

| Parameter | Units | 30170669001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Arsenic | mg/L | ND | ND | | |
| Barium | mg/L | ND | .27J | | |
| Cadmium | mg/L | ND | .00096J | | |
| Chromium | mg/L | ND | .0022J | | |
| Lead | mg/L | ND | .011J | | |
| Selenium | mg/L | ND | .01J | | |
| Silver | mg/L | ND | ND | | |

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## QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: MPRP/3681 Analysis Method: EPA 9023
QC Batch Method: EPA 9023 Analysis Description: 9023 Extractable Organic Halides EOX
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 194616 Matrix: Solid
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Extractable Organic Halogens | mg/kg | ND | 51.0 | 01/27/16 10:59 | |

LABORATORY CONTROL SAMPLE: 194617

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Extractable Organic Halogens | mg/kg | 1020 | 828 | 81 | 80-120 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 194618 194619

| Parameter | Units | 30170621001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Extractable Organic Halogens | mg/kg | ND | 946 | 1010 | 801 | 868 | 83 | 85 | 75-125 | 8 | |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: MSV/26737 Analysis Method: EPA 8260B
QC Batch Method: EPA 8260B Analysis Description: 8260B MSV TCLP
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013081 Matrix: Solid
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 1,1-Dichloroethene | ug/L | ND | 50.0 | 01/26/16 00:48 | |
| 1,2-Dichloroethane | ug/L | ND | 50.0 | 01/26/16 00:48 | |
| 2-Butanone (MEK) | ug/L | ND | 5000 | 01/26/16 00:48 | |
| Benzene | ug/L | ND | 50.0 | 01/26/16 00:48 | |
| Carbon tetrachloride | ug/L | ND | 50.0 | 01/26/16 00:48 | |
| Chlorobenzene | ug/L | ND | 1000 | 01/26/16 00:48 | |
| Chloroform | ug/L | ND | 500 | 01/26/16 00:48 | |
| Tetrachloroethene | ug/L | ND | 50.0 | 01/26/16 00:48 | |
| Trichloroethene | ug/L | ND | 50.0 | 01/26/16 00:48 | |
| Vinyl chloride | ug/L | ND | 50.0 | 01/26/16 00:48 | |
| 1,2-Dichloroethane-d4 (S) | % | 105 | 77-126 | 01/26/16 00:48 | |
| 4-Bromofluorobenzene (S) | % | 95 | 81-119 | 01/26/16 00:48 | |
| Dibromofluoromethane (S) | % | 100 | 70-130 | 01/26/16 00:48 | |
| Toluene-d8 (S) | % | 93 | 84-115 | 01/26/16 00:48 | |

LABORATORY CONTROL SAMPLE: 1015806

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 1,1-Dichloroethene | ug/L | 20 | 18.0 | 90 | 59-133 | |
| 1,2-Dichloroethane | ug/L | 20 | 17.9 | 89 | 66-123 | |
| 2-Butanone (MEK) | ug/L | 20 | 12J | 60 | 57-126 | |
| Benzene | ug/L | 20 | 18.5 | 92 | 69-115 | |
| Carbon tetrachloride | ug/L | 20 | 20.8 | 104 | 65-138 | |
| Chlorobenzene | ug/L | 20 | 18.8J | 94 | 69-120 | |
| Chloroform | ug/L | 20 | 18.8J | 94 | 67-123 | |
| Tetrachloroethene | ug/L | 20 | 19.8 | 99 | 62-122 | |
| Trichloroethene | ug/L | 20 | 18.9 | 94 | 61-126 | |
| Vinyl chloride | ug/L | 20 | 16.5 | 82 | 58-127 | |
| 1,2-Dichloroethane-d4 (S) | % | | | 104 | 77-126 | |
| 4-Bromofluorobenzene (S) | % | | | 96 | 81-119 | |
| Dibromofluoromethane (S) | % | | | 103 | 70-130 | |
| Toluene-d8 (S) | % | | | 94 | 84-115 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 1015807 1015808

| Parameter | Units | 30170621001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,1-Dichloroethene | ug/L | ND | 200 | 200 | 146 | 138 | 73 | 69 | 48-141 | 5 | |

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## QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 1015807 1015808

| Parameter | Units | 30170621001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,2-Dichloroethane | ug/L | ND | 200 | 200 | 159 | 154 | 80 | 77 | 58-123 | 3 | |
| 2-Butanone (MEK) | ug/L | ND | 200 | 200 | 321J | 331J | 79 | 84 | 43-128 | | |
| Benzene | ug/L | ND | 200 | 200 | 155 | 148 | 78 | 74 | 63-123 | 5 | |
| Carbon tetrachloride | ug/L | ND | 200 | 200 | 148 | 137 | 74 | 69 | 44-155 | 8 | |
| Chlorobenzene | ug/L | ND | 200 | 200 | 120J | 111J | 60 | 56 | 57-121 | | M1 |
| Chloroform | ug/L | ND | 200 | 200 | 154J | 147J | 77 | 73 | 56-132 | | |
| Tetrachloroethene | ug/L | ND | 200 | 200 | 112 | 109 | 56 | 55 | 53-125 | 2 | |
| Trichloroethene | ug/L | ND | 200 | 200 | 128 | 123 | 64 | 61 | 50-127 | 4 | |
| Vinyl chloride | ug/L | ND | 200 | 200 | 154 | 125 | 77 | 62 | 54-149 | 21 | |
| 1,2-Dichloroethane-d4 (S) | % | | | | | | 100 | 108 | 77-126 | | |
| 4-Bromofluorobenzene (S) | % | | | | | | 97 | 98 | 81-119 | | |
| Dibromofluoromethane (S) | % | | | | | | 104 | 104 | 70-130 | | |
| Toluene-d8 (S) | % | | | | | | 92 | 99 | 84-115 | | |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: OEXT/26876 Analysis Method: EPA 8015B
QC Batch Method: EPA 3546 Analysis Description: EPA 8015 TPH
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013992 Matrix: Solid
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| TPH (C10-C28) | mg/kg | ND | 6.7 | 01/22/16 00:34 | |
| o-Terphenyl (S) | % | 71 | 45-103 | 01/22/16 00:34 | |

LABORATORY CONTROL SAMPLE: 1013993

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| TPH (C10-C28) | mg/kg | 66.7 | 55.0 | 83 | 64-109 | |
| o-Terphenyl (S) | % | | | 79 | 45-103 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 1013994 1013995

| Parameter | Units | 30170701001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| TPH (C10-C28) | mg/kg | 42.6 | 66.5 | 66.1 | 98.8 | 93.6 | 84 | 77 | 10-175 | 5 | |
| o-Terphenyl (S) | % | | | | | | 84 | 91 | 45-103 | | |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: OEXT/26856 — Analysis Method: EPA 8081A
QC Batch Method: EPA 3510C — Analysis Description: 8081 GCS TCLP Pesticides
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013291 — Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Chlordane (Technical) | ug/L | ND | 10.0 | 01/21/16 13:57 | 1c |
| Endrin | ug/L | ND | 1.0 | 01/21/16 13:57 | 1c |
| gamma-BHC (Lindane) | ug/L | ND | 10.0 | 01/21/16 13:57 | 1c |
| Heptachlor epoxide | ug/L | ND | 0.50 | 01/21/16 13:57 | 1c |
| Methoxychlor | ug/L | ND | 100 | 01/21/16 13:57 | 1c |
| Toxaphene | ug/L | ND | 50.0 | 01/21/16 13:57 | 1c |
| Decachlorobiphenyl (S) | % | 38 | 15-125 | 01/21/16 13:57 | 1c,SS |
| Tetrachloro-m-xylene (S) | % | 41 | 14-136 | 01/21/16 13:57 | 1c |

METHOD BLANK: 1013295 — Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Chlordane (Technical) | ug/L | ND | 10.0 | 01/21/16 16:14 | 1c |
| Endrin | ug/L | ND | 1.0 | 01/21/16 16:14 | 1c |
| gamma-BHC (Lindane) | ug/L | ND | 10.0 | 01/21/16 16:14 | 1c |
| Heptachlor epoxide | ug/L | ND | 0.50 | 01/21/16 16:14 | 1c |
| Methoxychlor | ug/L | ND | 100 | 01/21/16 16:14 | 1c |
| Toxaphene | ug/L | ND | 50.0 | 01/21/16 16:14 | 1c |
| Decachlorobiphenyl (S) | % | 19 | 15-125 | 01/21/16 16:14 | 1c,SS |
| Tetrachloro-m-xylene (S) | % | 52 | 14-136 | 01/21/16 16:14 | 1c |

METHOD BLANK: 1013296 — Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Chlordane (Technical) | ug/L | ND | 10.0 | 01/21/16 18:04 | 1c |
| Endrin | ug/L | ND | 1.0 | 01/21/16 18:04 | 1c |
| gamma-BHC (Lindane) | ug/L | ND | 10.0 | 01/21/16 18:04 | 1c |
| Heptachlor epoxide | ug/L | ND | 0.50 | 01/21/16 18:04 | 1c |
| Methoxychlor | ug/L | ND | 100 | 01/21/16 18:04 | 1c |
| Toxaphene | ug/L | ND | 50.0 | 01/21/16 18:04 | 1c |
| Decachlorobiphenyl (S) | % | 26 | 15-125 | 01/21/16 18:04 | 1c,SS |
| Tetrachloro-m-xylene (S) | % | 50 | 14-136 | 01/21/16 18:04 | 1c |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

LABORATORY CONTROL SAMPLE: 1013292

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| Endrin | ug/L | 1.6 | 1.4 | 85 | 64-110 | 1c |
| gamma-BHC (Lindane) | ug/L | .8 | .64J | 80 | 61-109 | 1c |
| Heptachlor epoxide | ug/L | .8 | 0.61 | 76 | 56-104 | 1c |
| Methoxychlor | ug/L | 8 | 6.7J | 84 | 61-108 | 1c |
| Decachlorobiphenyl (S) | % | | | 58 | 15-125 | 1c,SS |
| Tetrachloro-m-xylene (S) | % | | | 55 | 14-136 | 1c |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 1013293 1013294

| Parameter | Units | 30170455002 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Endrin | ug/L | ND | 1.6 | 1.6 | 1.4 | 1.4 | 88 | 87 | 64-110 | 1 | 1c |
| gamma-BHC (Lindane) | ug/L | ND | .8 | .8 | .65J | .63J | 81 | 79 | 61-109 | | 1c |
| Heptachlor epoxide | ug/L | ND | .8 | .8 | 0.58 | 0.56 | 70 | 68 | 56-104 | 2 | 1c |
| Methoxychlor | ug/L | ND | 8 | 8 | 7.1J | 7.2J | 89 | 91 | 61-108 | | 1c |
| Decachlorobiphenyl (S) | % | | | | | | 40 | 44 | 15-125 | | 1c,SS |
| Tetrachloro-m-xylene (S) | % | | | | | | 60 | 60 | 14-136 | | 1c |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: OEXT/26850    Analysis Method: EPA 8082
QC Batch Method: EPA 3546    Analysis Description: 8082 GCS PCB
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013126    Matrix: Solid
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| PCB-1016 (Aroclor 1016) | ug/kg | ND | 16.7 | 01/20/16 12:39 | |
| PCB-1221 (Aroclor 1221) | ug/kg | ND | 16.7 | 01/20/16 12:39 | |
| PCB-1232 (Aroclor 1232) | ug/kg | ND | 16.7 | 01/20/16 12:39 | |
| PCB-1242 (Aroclor 1242) | ug/kg | ND | 16.7 | 01/20/16 12:39 | |
| PCB-1248 (Aroclor 1248) | ug/kg | ND | 16.7 | 01/20/16 12:39 | |
| PCB-1254 (Aroclor 1254) | ug/kg | ND | 16.7 | 01/20/16 12:39 | |
| PCB-1260 (Aroclor 1260) | ug/kg | ND | 16.7 | 01/20/16 12:39 | |
| Decachlorobiphenyl (S) | % | 53 | 10-115 | 01/20/16 12:39 | CL |
| Tetrachloro-m-xylene (S) | % | 72 | 30-107 | 01/20/16 12:39 | |

LABORATORY CONTROL SAMPLE: 1013127

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| PCB-1016 (Aroclor 1016) | ug/kg | 167 | 125 | 75 | 40-100 | |
| PCB-1260 (Aroclor 1260) | ug/kg | 167 | 134 | 80 | 41-109 | 4c |
| Decachlorobiphenyl (S) | % | | | 60 | 10-115 | CL |
| Tetrachloro-m-xylene (S) | % | | | 73 | 30-107 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 1013128    1013129

| Parameter | Units | 30170691001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| PCB-1016 (Aroclor 1016) | ug/kg | ND | 827 | 847 | 655 | 720 | 79 | 85 | 40-100 | 9 | |
| PCB-1260 (Aroclor 1260) | ug/kg | ND | 827 | 847 | 672 | 649 | 81 | 77 | 41-109 | 4 | |
| Decachlorobiphenyl (S) | % | | | | | | 78 | 81 | 10-115 | | CH |
| Tetrachloro-m-xylene (S) | % | | | | | | 55 | 57 | 30-107 | | CH |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: OEXT/7866 | Analysis Method: EPA 8151
QC Batch Method: EPA 3535A | Analysis Description: 8151 GCS TCLP Herbicides
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 193033 | Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 2,4,5-TP (Silvex) | mg/L | ND | 0.020 | 01/22/16 13:42 | |
| 2,4-D | mg/L | ND | 0.020 | 01/22/16 13:42 | |
| 2,4-DCAA (S) | %. | 96 | 10-166 | 01/22/16 13:42 | |
| 2,4-DCAA (S) | %. | 102 | 10-166 | 01/22/16 13:42 | |

LABORATORY CONTROL SAMPLE: 193034

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 2,4,5-TP (Silvex) | mg/L | .04 | 0.040 | 99 | 22-158 | |
| 2,4-D | mg/L | .04 | 0.036 | 91 | 10-151 | |
| 2,4-DCAA (S) | %. | | | 108 | 10-166 | |
| 2,4-DCAA (S) | %. | | | 105 | 10-166 | |

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 193035 193036

| Parameter | Units | 30170621001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 2,4,5-TP (Silvex) | mg/L | ND | .04 | .04 | 0.033 | 0.023 | 80 | 55 | 16-164 | 35 | R1 |
| 2,4-D | mg/L | ND | .04 | .04 | 0.031 | 0.022 | 77 | 54 | 10-160 | 35 | R1 |
| 2,4-DCAA (S) | %. | | | | | | 90 | 65 | 10-166 | | |
| 2,4-DCAA (S) | %. | | | | | | 87 | 63 | 10-166 | | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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(724)850-5600

# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: OEXT/26851 — Analysis Method: EPA 8270C
QC Batch Method: EPA 3510C — Analysis Description: 8270 TCLP MSSV
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013170 — Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| 1,4-Dichlorobenzene | ug/L | ND | 500 | 01/19/16 20:48 | |
| 2,4,5-Trichlorophenol | ug/L | ND | 5000 | 01/19/16 20:48 | |
| 2,4,6-Trichlorophenol | ug/L | ND | 100 | 01/19/16 20:48 | |
| 2,4-Dinitrotoluene | ug/L | ND | 100 | 01/19/16 20:48 | |
| 2-Methylphenol(o-Cresol) | ug/L | ND | 2000 | 01/19/16 20:48 | |
| 3&4-Methylphenol(m&p Cresol) | ug/L | ND | 2000 | 01/19/16 20:48 | |
| Hexachloro-1,3-butadiene | ug/L | ND | 100 | 01/19/16 20:48 | |
| Hexachlorobenzene | ug/L | ND | 100 | 01/19/16 20:48 | |
| Hexachloroethane | ug/L | ND | 500 | 01/19/16 20:48 | |
| Nitrobenzene | ug/L | ND | 100 | 01/19/16 20:48 | |
| Pentachlorophenol | ug/L | ND | 5000 | 01/19/16 20:48 | |
| Pyridine | ug/L | ND | 500 | 01/19/16 20:48 | |
| 2,4,6-Tribromophenol (S) | % | 96 | 34-134 | 01/19/16 20:48 | |
| 2-Fluorobiphenyl (S) | % | 88 | 34-113 | 01/19/16 20:48 | |
| 2-Fluorophenol (S) | % | 61 | 19-70 | 01/19/16 20:48 | |
| Nitrobenzene-d5 (S) | % | 85 | 22-128 | 01/19/16 20:48 | |
| Phenol-d6 (S) | % | 41 | 14-49 | 01/19/16 20:48 | |
| Terphenyl-d14 (S) | % | 94 | 35-150 | 01/19/16 20:48 | |

LABORATORY CONTROL SAMPLE: 1013171

| Parameter | Units | Spike Conc. | LCS Result | LCS % Rec | % Rec Limits | Qualifiers |
|---|---|---|---|---|---|---|
| 1,4-Dichlorobenzene | ug/L | 500 | 396J | 79 | 34-91 | |
| 2,4,5-Trichlorophenol | ug/L | 500 | 442J | 88 | 23-160 | |
| 2,4,6-Trichlorophenol | ug/L | 500 | 463 | 93 | 51-127 | |
| 2,4-Dinitrotoluene | ug/L | 500 | 385 | 77 | 46-107 | |
| 2-Methylphenol(o-Cresol) | ug/L | 500 | 367J | 73 | 32-116 | |
| 3&4-Methylphenol(m&p Cresol) | ug/L | 1000 | 750J | 75 | 30-103 | |
| Hexachloro-1,3-butadiene | ug/L | 500 | 428 | 86 | 36-117 | |
| Hexachlorobenzene | ug/L | 500 | 315 | 63 | 53-128 | |
| Hexachloroethane | ug/L | 500 | 398J | 80 | 26-110 | |
| Nitrobenzene | ug/L | 500 | 448 | 90 | 26-130 | |
| Pentachlorophenol | ug/L | 500 | 434J | 87 | 28-131 | |
| Pyridine | ug/L | 500 | 374J | 75 | 10-175 | |
| 2,4,6-Tribromophenol (S) | % | | | 96 | 34-134 | |
| 2-Fluorobiphenyl (S) | % | | | 81 | 34-113 | |
| 2-Fluorophenol (S) | % | | | 60 | 19-70 | |
| Nitrobenzene-d5 (S) | % | | | 84 | 22-128 | |
| Phenol-d6 (S) | % | | | 40 | 14-49 | |
| Terphenyl-d14 (S) | % | | | 91 | 35-150 | |

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

MATRIX SPIKE & MATRIX SPIKE DUPLICATE: 1013172 1013173

| Parameter | Units | 30170669001 Result | MS Spike Conc. | MSD Spike Conc. | MS Result | MSD Result | MS % Rec | MSD % Rec | % Rec Limits | RPD | Qual |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1,4-Dichlorobenzene | ug/L | ND | 500 | 500 | 358J | 394J | 72 | 79 | 34-91 | | |
| 2,4,5-Trichlorophenol | ug/L | ND | 500 | 500 | 395J | 448J | 79 | 90 | 23-160 | | |
| 2,4,6-Trichlorophenol | ug/L | ND | 500 | 500 | 400 | 425 | 80 | 85 | 51-127 | 6 | |
| 2,4-Dinitrotoluene | ug/L | ND | 500 | 500 | 334 | 382 | 67 | 76 | 46-107 | 13 | |
| 2-Methylphenol(o-Cresol) | ug/L | ND | 500 | 500 | 338J | 370J | 68 | 74 | 32-116 | | |
| 3&4-Methylphenol(m&p Cresol) | ug/L | ND | 1000 | 1000 | 689J | 736J | 69 | 74 | 30-103 | | |
| Hexachloro-1,3-butadiene | ug/L | ND | 500 | 500 | 404 | 471 | 81 | 94 | 36-117 | 15 | |
| Hexachlorobenzene | ug/L | ND | 500 | 500 | 313 | 335 | 63 | 67 | 53-128 | 7 | |
| Hexachloroethane | ug/L | ND | 500 | 500 | 380J | 378J | 76 | 76 | 26-110 | | |
| Nitrobenzene | ug/L | ND | 500 | 500 | 401 | 451 | 80 | 90 | 26-130 | 12 | |
| Pentachlorophenol | ug/L | ND | 500 | 500 | 337J | 418J | 67 | 84 | 28-131 | | |
| Pyridine | ug/L | ND | 500 | 500 | 334J | 323J | 67 | 65 | 10-175 | | |
| 2,4,6-Tribromophenol (S) | % | | | | | | 89 | 94 | 34-134 | | |
| 2-Fluorobiphenyl (S) | % | | | | | | 76 | 87 | 34-113 | | |
| 2-Fluorophenol (S) | % | | | | | | 56 | 54 | 19-70 | | |
| Nitrobenzene-d5 (S) | % | | | | | | 77 | 86 | 22-128 | | |
| Phenol-d6 (S) | % | | | | | | 40 | 39 | 14-49 | | |
| Terphenyl-d14 (S) | % | | | | | | 88 | 93 | 35-150 | | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: PMST/5883
Analysis Method: Dry Weight
QC Batch Method: Dry Weight
Analysis Description: Dry Weight/Percent Moisture
Associated Lab Samples: 30170621001, 30170621002, 30170621003

SAMPLE DUPLICATE: 1014464

| Parameter | Units | 30170035001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Percent Moisture | % | 83.1 | 83.3 | 0 | |

SAMPLE DUPLICATE: 1014465

| Parameter | Units | 30170516001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Percent Moisture | % | 1.6 | 1.4 | 14 | |

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## QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: WET/31785 Analysis Method: EPA 1010
QC Batch Method: EPA 1010 Analysis Description: 1010 Flash Point, Closed Cup
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1014941 Matrix: Water
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Flashpoint | deg F | >200 | 60.0 | 01/22/16 16:58 | |

SAMPLE DUPLICATE: 1014942

| Parameter | Units | 30169962001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Flashpoint | deg F | >200 | >200 | | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: WET/31714 | Analysis Method: EPA 9045C
QC Batch Method: EPA 9045C | Analysis Description: 9045 pH
Associated Lab Samples: 30170621001, 30170621002, 30170621003

SAMPLE DUPLICATE: 1012477

| Parameter | Units | 30170692001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| pH at 25 Degrees C | Std. Units | 7.5 | 7.5 | 0 | |

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

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# QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: WETA/22416
Analysis Method: SW-846 7.3.3.2
QC Batch Method: SW-846 7.3.3.2
Analysis Description: 733C Reactive Cyanide
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013142
Matrix: Solid
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Cyanide, Reactive | mg/kg | ND | 0.99 | 01/19/16 23:33 | |

SAMPLE DUPLICATE: 1013143

| Parameter | Units | 30170778001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Cyanide, Reactive | mg/kg | ND | ND | | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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## QUALITY CONTROL DATA

Project: Transportation Drive 1546122
Pace Project No.: 30170621

QC Batch: WETA/22417 — Analysis Method: SW-846 7.3.4.2
QC Batch Method: SW-846 7.3.4.2 — Analysis Description: 734S Reactive Sulfide
Associated Lab Samples: 30170621001, 30170621002, 30170621003

METHOD BLANK: 1013144 — Matrix: Solid
Associated Lab Samples: 30170621001, 30170621002, 30170621003

| Parameter | Units | Blank Result | Reporting Limit | Analyzed | Qualifiers |
|---|---|---|---|---|---|
| Sulfide, Reactive | mg/kg | ND | 9.9 | 01/19/16 16:25 | |

SAMPLE DUPLICATE: 1013145

| Parameter | Units | 30170778001 Result | Dup Result | RPD | Qualifiers |
|---|---|---|---|---|---|
| Sulfide, Reactive | mg/kg | ND | ND | | |

**Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.**

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# QUALIFIERS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

## DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

## LABORATORIES

| | |
|---|---|
| PASI-N | Pace Analytical Services - New Orleans |
| PASI-PA | Pace Analytical Services - Greensburg |

## BATCH QUALIFIERS

Batch: GCSV/9103

| | |
|---|---|
| [1] | A retention time shift occurred during the analytical sequence such that analytes fell outside of their respective retention time windows in the CCVs. QC samples and calibration standards were used to aid in analyte identification. |

## ANALYTE QUALIFIERS

| | |
|---|---|
| 1c | A retention time shift occurred during the analytical sequence such that analytes fell outside of their respective retention time windows in the CCVs. QC samples and calibration standards were used to aid in analyte identification. |
| 2c | The result for this analyte is reported from the front column due to high response in the closing CCV on the rear column. |
| 3c | The result for this analyte is reported from the rear column due to high response in the closing CCV on the front column. |
| 4c | The result is reported from the rear column due to low response for this analyte in the closing CCV on the front column. |
| CH | The continuing calibration for this compound is outside of Pace Analytical acceptance limits. The results may be biased high. |
| CL | The continuing calibration for this compound is outside of Pace Analytical acceptance limits. The results may be biased low. |
| IS | The internal standard response is below criteria. Results may be biased high. |
| M1 | Matrix spike recovery exceeded QC limits. Batch accepted based on laboratory control sample (LCS) recovery. |
| R1 | RPD value was outside control limits. |
| S3 | Surrogate recovery exceeded laboratory control limits. Analyte presence below reporting limits in associated samples. Results unaffected by high bias. |

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## QUALIFIERS

Project: Transportation Drive 1546122
Pace Project No.: 30170621

### ANALYTE QUALIFIERS

| | |
|---|---|
| S4 | Surrogate recovery not evaluated against control limits due to sample dilution. |
| SS | This analyte did not meet the secondary source verification criteria for the initial calibration. The reported result should be considered an estimated value. |

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## QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Transportation Drive 1546122
Pace Project No.: 30170621

| Lab ID | Sample ID | QC Batch Method | QC Batch | Analytical Method | Analytical Batch |
|---|---|---|---|---|---|
| **30170621001** | **Schiavo SP-02-A** | EPA 3546 | OEXT/26876 | EPA 8015B | GCSV/9106 |
| **30170621002** | **Schiavo SP-02-B** | EPA 3546 | OEXT/26876 | EPA 8015B | GCSV/9106 |
| **30170621003** | **Schiavo SP-02-C** | EPA 3546 | OEXT/26876 | EPA 8015B | GCSV/9106 |
| **30170621001** | **Schiavo SP-02-A** | EPA 3510C | OEXT/26856 | EPA 8081A | GCSV/9103 |
| **30170621002** | **Schiavo SP-02-B** | EPA 3510C | OEXT/26856 | EPA 8081A | GCSV/9103 |
| **30170621003** | **Schiavo SP-02-C** | EPA 3510C | OEXT/26856 | EPA 8081A | GCSV/9103 |
| **30170621001** | **Schiavo SP-02-A** | EPA 3546 | OEXT/26850 | EPA 8082 | GCSV/9097 |
| **30170621002** | **Schiavo SP-02-B** | EPA 3546 | OEXT/26850 | EPA 8082 | GCSV/9097 |
| **30170621003** | **Schiavo SP-02-C** | EPA 3546 | OEXT/26850 | EPA 8082 | GCSV/9097 |
| **30170621001** | **Schiavo SP-02-A** | EPA 3535A | OEXT/7866 | EPA 8151 | GCSV/5714 |
| **30170621002** | **Schiavo SP-02-B** | EPA 3535A | OEXT/7866 | EPA 8151 | GCSV/5714 |
| **30170621003** | **Schiavo SP-02-C** | EPA 3535A | OEXT/7866 | EPA 8151 | GCSV/5714 |
| **30170621001** | **Schiavo SP-02-A** | EPA 3005A | MPRP/17380 | EPA 6010B | ICP/16499 |
| **30170621002** | **Schiavo SP-02-B** | EPA 3005A | MPRP/17380 | EPA 6010B | ICP/16499 |
| **30170621003** | **Schiavo SP-02-C** | EPA 3005A | MPRP/17380 | EPA 6010B | ICP/16499 |
| **30170621001** | **Schiavo SP-02-A** | EPA 7470A | MERP/7283 | EPA 7470A | MERC/6964 |
| **30170621002** | **Schiavo SP-02-B** | EPA 7470A | MERP/7283 | EPA 7470A | MERC/6964 |
| **30170621003** | **Schiavo SP-02-C** | EPA 7470A | MERP/7283 | EPA 7470A | MERC/6964 |
| **30170621001** | **Schiavo SP-02-A** | EPA 9023 | MPRP/3681 | EPA 9023 | MPRP/3683 |
| **30170621002** | **Schiavo SP-02-B** | EPA 9023 | MPRP/3681 | EPA 9023 | MPRP/3683 |
| **30170621003** | **Schiavo SP-02-C** | EPA 9023 | MPRP/3681 | EPA 9023 | MPRP/3683 |
| **30170621001** | **Schiavo SP-02-A** | EPA 3510C | OEXT/26851 | EPA 8270C | MSSV/8753 |
| **30170621002** | **Schiavo SP-02-B** | EPA 3510C | OEXT/26851 | EPA 8270C | MSSV/8753 |
| **30170621003** | **Schiavo SP-02-C** | EPA 3510C | OEXT/26851 | EPA 8270C | MSSV/8753 |
| **30170621001** | **Schiavo SP-02-A** | EPA 8260B | MSV/26737 | | |
| **30170621002** | **Schiavo SP-02-B** | EPA 8260B | MSV/26737 | | |
| **30170621003** | **Schiavo SP-02-C** | EPA 8260B | MSV/26737 | | |
| **30170621001** | **Schiavo SP-02-A** | Dry Weight | PMST/5883 | | |
| **30170621002** | **Schiavo SP-02-B** | Dry Weight | PMST/5883 | | |
| **30170621003** | **Schiavo SP-02-C** | Dry Weight | PMST/5883 | | |
| **30170621001** | **Schiavo SP-02-A** | EPA 1010 | WET/31785 | | |
| **30170621002** | **Schiavo SP-02-B** | EPA 1010 | WET/31785 | | |
| **30170621003** | **Schiavo SP-02-C** | EPA 1010 | WET/31785 | | |
| **30170621001** | **Schiavo SP-02-A** | EPA 9045C | WET/31714 | | |
| **30170621002** | **Schiavo SP-02-B** | EPA 9045C | WET/31714 | | |
| **30170621003** | **Schiavo SP-02-C** | EPA 9045C | WET/31714 | | |
| **30170621001** | **Schiavo SP-02-A** | SW-846 7.3.3.2 | WETA/22416 | | |
| **30170621002** | **Schiavo SP-02-B** | SW-846 7.3.3.2 | WETA/22416 | | |
| **30170621003** | **Schiavo SP-02-C** | SW-846 7.3.3.2 | WETA/22416 | | |
| **30170621001** | **Schiavo SP-02-A** | SW-846 7.3.4.2 | WETA/22417 | | |
| **30170621002** | **Schiavo SP-02-B** | SW-846 7.3.4.2 | WETA/22417 | | |
| **30170621003** | **Schiavo SP-02-C** | SW-846 7.3.4.2 | WETA/22417 | | |

# CHAIN-OF-CUSTODY / Analytical Request Document

The Chain-of-Custody is a LEGAL DOCUMENT. All relevant fields must be completed accurately.

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Page: of

1973557

**Section A**
**Required** Client Information:

| | |
|---|---|
| Company: | North Star |
| Address: | 55 Progress Place<br>Jackson, NJ |
| Email To: | GGommer@northstar.com |
| Phone: | 213-339-3341 Fax: |
| Requested Due Date/TAT: | Normal |

**Section B**
**Required** Project Information:

| | |
|---|---|
| Report To: | Gene Gommer |
| Copy To: | same |
| Purchase Order No.: | |
| Project Name: | Transportation Drive |
| Project Number: | 1546122 |

**Section C**
Invoice Information:

| | |
|---|---|
| Attention: | Gene Gommer |
| Company Name: | Northstar |
| Address: | Same |
| Pace Quote Reference: | 120021884 |
| Pace Project Manager: | Samantha |
| Pace Profile #: | |

**REGULATORY AGENCY**

- [ ] NPDES
- [ ] GROUND WATER
- [ ] DRINKING WATER
- [ ] UST
- [x] RCRA
- [ ] OTHER ______

Site Location STATE: PA

**Section D**
**Required** Client Information

**SAMPLE ID**
(A-Z, 0-9 / ,-)
Sample IDs MUST BE UNIQUE

| Matrix Codes MATRIX | CODE |
|---|---|
| Drinking Water | DW |
| Water | WT |
| Waste Water | WW |
| Product | P |
| Soil/Solid | SL |
| Oil | OL |
| Wipe | WP |
| Air | AR |
| Tissue | TS |
| Other | OT |

| ITEM # | SAMPLE ID | MATRIX CODE (see valid codes to left) | SAMPLE TYPE (G=GRAB C=COMP) | COLLECTED COMPOSITE START DATE | COMPOSITE START TIME | COMPOSITE END/GRAB DATE | COMPOSITE END/GRAB TIME | SAMPLE TEMP AT COLLECTION | # OF CONTAINERS |
|---|---|---|---|---|---|---|---|---|---|
| 1 | Schiavo SP-02-A | SL | C | 1/13/16 | 0930 | 1/13 | 1000 | | 1 |
| 2 | Schiavo SP-02-B | SL | C | 1/13/16 | 1015 | 1/13 | 1045 | | 1 |
| 3 | Schiavo SP-02-C | SL | C | 1/13/16 | 1100 | 1/13 | 1145 | | 1 |
| 4 | | | | | | | | | |
| 5 | | | | | | | | | |
| 6 | | | | | | | | | |
| 7 | | | | | | | | | |
| 8 | | | | | | | | | |
| 9 | | | | | | | | | |
| 10 | | | | | | | | | |
| 11 | | | | | | | | | |
| 12 | | | | | | | | | |

Preservatives: Unpreserved, $H_2SO_4$, $HNO_3$, HCl, NaOH, $Na_2S_2O_3$, Methanol, Other (none marked)

Requested Analysis Filtered (Y/N)

| ITEM # | Analysis Test: 8082 8015B 9015A | 8081/6010B, 7470A | 8270C 8260B | Dry weight 1010 | 9045C/SW-846 | 7.2.3/SW-846 | 7.3.4.2/9023 | Residual Chlorine (Y/N) | Pace Project No./ Lab I.D. |
|---|---|---|---|---|---|---|---|---|---|
| 1 | X | X | X | X | X | X | X | | 001 |
| 2 | X | X | X | X | X | X | X | | 002 |
| 3 | X | X | X | X | X | X | X | | 003 |

30170621

| ADDITIONAL COMMENTS | RELINQUISHED BY / AFFILIATION | DATE | TIME | ACCEPTED BY / AFFILIATION | DATE | TIME |
|---|---|---|---|---|---|---|
| | [signature] | 1/14/16 | 1200 | [signature] | 1/14/16 | 1300 |
| | [signature] | 1/14/16 | 1600 | [signature] | 1/14/16 | 1600 |
| | [signature] | 1/14/16 | 1720 | [signature] Pace | 1.14.15 | 1920 |
| | [signature] | 1.14.16 | 2230 | [signature] Pace | 1-14-16 | 2230 |

SAMPLE CONDITIONS

| Temp in °C | Received on Ice (Y/N) | Custody Sealed Cooler (Y/N) | Samples Intact (Y/N) |
|---|---|---|---|
| 3.4 | Y | Y | Y |

ORIGINAL

**SAMPLER NAME AND SIGNATURE**

PRINT Name of SAMPLER:

SIGNATURE of SAMPLER: DATE Signed (MM/DD/YY):

*Important Note: By signing this form you are accepting Pace's NET 30 day payment terms and agreeing to late charges of 1.5% per month for any invoices not paid within 30 days.

AR100565

F-ALL-Q-020rev.07, 15-May-2007

BCM
30170621

# Sample Condition Upon Receipt

Pace Analytical

Client Name: North Star    Project #

**Courier:** ☐ Fed Ex ☐ UPS ☐ USPS ☐ Client ☐ Commercial ☑ Pace Other ______
**Tracking #:** N/A
**Custody Seal on Cooler/Box Present:** ☑ yes ☐ no   Seals intact: ☑ yes ☐ no   **Biological Tissue is Frozen:** Yes No
**Packing Material:** Bubble Wrap ____ Bubble Bags ✓ None ____ Other ______
**Thermometer Used** 7   **Type of Ice:** Wet Blue None   ☑ Samples on ice, cooling process has begun
**Cooler Temp.: Observed Temp.:** 3.0 °C **Correction Factor:** +0.4 °C **Final Temp:** 3.4 °C

Date and Initials of person examining contents: JL 1-15-16

Temp should be above freezing to 6°C

| | | | | Comments: |
|---|---|---|---|---|
| Chain of Custody Present: | ☑Yes | ☐No | ☐N/A | 1. |
| Chain of Custody Filled Out: | ☑Yes | ☐No | ☐N/A | 2. |
| Chain of Custody Relinquished: | ☑Yes | ☐No | ☐N/A | 3. |
| Sampler Name & Signature on COC: | ☐Yes | ☑No | ☐N/A | 4. |
| Samples Arrived within Hold Time: | ☑Yes | ☐No | ☐N/A | 5. |
| **Short Hold Time Analysis (<72hr):** | ☐Yes | ☑No | ☐N/A | 6. |
| **Rush Turn Around Time Requested:** | ~~☑Yes~~ | ☑No | ☐N/A | 7. JL 1/15/16 |
| Sufficient Volume: | ☑Yes | ☐No | ☐N/A | 8. |
| Correct Containers Used: | ☑Yes | ☐No | ☐N/A | 9. |
| -Pace Containers Used: | ☑Yes | ☐No | ☐N/A | |
| Containers Intact: | ☑Yes | ☐No | ☐N/A | 10. |
| Filtered volume received for Dissolved tests | ☐Yes | ☐No | ☑N/A | 11. |
| Sample Labels match COC: | ☐Yes | ☑No | ☐N/A | 12. "Schiavo" not on SB-02-C |
| -Includes date/time/ID/Analysis Matrix: SL | | | | |
| All containers needing preservation have been checked. | ☐Yes | ☐No | ☑N/A | 13. |
| All containers needing preservation are found to be in compliance with EPA recommendation. | ☐Yes | ☐No | ☑N/A | |
| exceptions: VOA, coliform, TOC, O&G, Phenols | ☐Yes | ☑No | | Initial when completed JL / Lot # of added preservative |
| Samples checked for dechlorination: | ☐Yes | ☐No | ☑N/A | 14. |
| Headspace in VOA Vials ( >6mm): | ☐Yes | ☐No | ☑N/A | 15. |
| Trip Blank Present: | ☐Yes | ☐No | ☑N/A | 16. |
| Trip Blank Custody Seals Present | ☐Yes | ☐No | ☑N/A | |
| Pace Trip Blank Lot # (if purchased): ______ | | | | |

**Client Notification/ Resolution:**   Field Data Required? Y / N

Person Contacted: ______ Date/Time: ______

Comments/ Resolution: ______

**Project Manager Review:** [signature]   **Date:** 1/19/16

Note: Whenever there is a discrepancy affecting North Carolina compliance samples, a copy of this form will be sent to the North Carolina DEHNR Certification Office ( i.e out of hold, incorrect preservative, out of temp, incorrect containers)

30170621

Pace Analytical

page 2

**Project Number:** ____________________

**Client Name:** North Star

| Item No. | Matrix Code | Glass Jar (120 / 250 / 500 / 1L) Amber | Soil kit (2 SB, 1M, soil jar) | Chemistry (250 / 500 / 1L) | Organics (1L) | Nutrient (250 / 500) | Phenolics (250 ml) | TOC (40 ml / 250 ml) | TOX (250 ml) | Total Metals | Dissolved Metals preserved Y N | O & G (1L) | TPH (1L) | VOA (40 ml 30 ml) | Cyanide (250 ml) | Sulfide (500 ml) | Bacteria (120 ml) | Wipes / swipe/ smear/ filter | Radchem Nalgene (125 / 250 / 500 / 1L) | Radchem Nalgene (1/2 gal. / 1 gal.L) | Cubitainer (500 ml / 4L) | Ziploc | Other | Other |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 001 | SL | 1 | | | | | | | | | | | | | | | | | | | | | | |
| 002 | SL | 1 | | | | | | | | | | | | | | | | | | | | | | |
| 003 | SL | 1 | | | | | | | | | | | | | | | | | | | | | | |
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SCURF Back (C016-4 15May2012).xls

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